**Subject:** Mathematics
**Class:** Class X (CBSE)

**CBSE CLASS 10 – WEEKLY TEST 05**
**Chapter 1: Real Numbers**

**Time:** 45 Minutes
**MM:** 20

**Student Name:** _____ **Roll No.:** _____ **Date:** _____

**GENERAL INSTRUCTIONS:**

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer (VSA) question of 2 marks.
4. **Section C** comprises 3 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 1 Case Study based question of 4 marks with internal subparts.
6. Use of calculators is strictly prohibited. All mathematical steps must be clearly shown.

## SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

**Q1.** If $p$ and $q$ are two co-prime natural numbers, then LCM$(p, q)$ is equal to: [1]
(a) 1 (b) $\frac{p}{q}$ (c) $p + q$ (d) $pq$

**Q2.** Given that HCF$(306, 657) = 9$, the value of LCM$(306, 657)$ is: [1]
(a) 22338 (b) 21338 (c) 22334 (d) 23238

**Q3.** Given that LCM$(91, 26) = 182$, the value of HCF$(91, 26)$ is: [1]
(a) 26 (b) 13 (c) 7 (d) 9

**Q4.** Which of the following expressions represents a rational number? [1]
(a) $(\sqrt{2} - \sqrt{3})(\sqrt{2} + \sqrt{3})$
(b) $(2 + \sqrt{3})^2$
(c) $\frac{\sqrt{3}}{2}$
(d) $3\sqrt{2}$

**Q5.** **Assertion (A):** The HCF and LCM of two natural numbers can be 18 and 380 respectively.
**Reason (R):** The LCM of two natural numbers is always completely divisible by their HCF. [1]
(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

## SECTION B: Very Short Answer Type (1 × 2 = 2 Marks)

**Q6.** Find the HCF and LCM of 612 and 1314 using the prime factorisation method. [2]

## SECTION C: Short Answer Type (3 × 3 = 9 Marks)

**Q7.** Three sets of English, Hindi, and Mathematics books have to be stacked in such a way that all the books are stored topic-wise and the height of each stack is the same. The number of English books is 96, Hindi books is 240, and Mathematics books is 336. Assuming the books are of identical thickness, determine the number of books in each stack and the total number of stacks. [3]

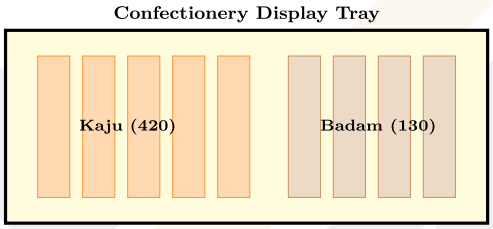
**Q8.** A circular field has a circumference of 360 km. Three cyclists start together and can cycle 48 km, 60 km, and 72 km a day around the field. When will they meet again at the starting point? [3]

**Q9.** Four bells toll together at 9:00 AM. They toll at intervals of 6, 8, 12, and 18 seconds respectively. How many times will they toll together in the next 1 hour (excluding the toll at 9:00 AM)? [3]

## SECTION D: Case Study Based Question (1 × 4 = 4 Marks)

**Q10. CASE STUDY: CASE STUDY 05: CONFECTIONERY SWEET SHOP TRAY PACKING** [4 Marks]

A traditional sweet manufacturer prepares a premier batch of artisanal sweets for a festive order. The batch consists of 420 pieces of Kaju Barfi and 130 pieces of Badam Barfi. The packaging team needs to stack them neatly on display trays such that each stack has the same number of pieces, each stack consists of only one variety of barfi, and the stacks occupy the minimum possible surface area of each tray.




(i) Find the maximum number of barfis that can be placed in each stack to minimize the tray area. [2]

(ii) Calculate the total number of stacks formed on the trays. [1]

(iii) Express 420 in its canonical prime factorisation form. [1]

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